

# Deepak Mallampati

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Kent, OH

## PROFESSIONAL SUMMARY

Applied AI engineer shipping production LLM systems end to end: healthcare RAG (~35% retrieval precision, >90% grounded alignment), agentic workflows (>30% hallucination reduction), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Shipped Manga Lens solo on the Chrome Web Store with four AI vision providers under per-provider failure isolation. Master's in Computer Science, Kent State. Ready to ship production AI systems from day one. Targeting AI Platform / MLOps Engineer (GPU Inference) roles (Spellbrush and similar).

## SKILLS

Applied AI / LLMs • RAG + Grounding • Agentic Workflows • Schema Contracts • Eval Harnesses • Python / FastAPI • TypeScript / React • Docker / CI-CD • Multimodal Vision • Predictive ML

**Languages:** Python, TypeScript, JavaScript, SQL, Java, C++

**AI / ML:** PyTorch, scikit-learn, XGBoost, LangChain, LangGraph, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, YOLOv8

**LLM / GenAI:** RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines (RAGAS, LangSmith, MLflow), grounding, embeddings, vector search

**Backend:** FastAPI, Flask, Node.js, REST, WebSockets, Docker, Kubernetes

**Frontend:** React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

**Data:** PostgreSQL + pgvector, Pinecone, Weaviate, FAISS, MongoDB, Redis, Pandas, NumPy

**DevOps:** Docker, Kubernetes, Jenkins, GitHub Actions, Grafana, Prometheus, AWS, Azure, GCP

**Domains:** Healthcare AI (HIPAA), Banking-Regulated AI, Applied AI, Enterprise AI, Multi-Agent Systems

## PROFESSIONAL EXPERIENCE

### Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built Retrieval-Augmented Generation (RAG) for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~35%, reduced irrelevant retrieval >30%, response alignment >90%.
- Developed agentic LLM workflows for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~25%, hallucinations cut >30%.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +15-20% on high-risk categories.
- Packaged ML/LLM inference as FastAPI / Flask services on Docker with structured logging; post-deploy defects ~30%.
- EHR preprocessing (Pandas, NumPy); dataset reliability >98%, downstream instability -40%.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails.

### Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized T-SQL stored procedures on a compliance-sensitive ERP; query time -20%, retrieval latency -25%.
- Built Jenkins CI/CD pipelines; deploy errors -30%, release cycle -35 to -40%.
- Designed performance dashboards with SQL Server DMVs + Grafana; incident recurrence -25%.
- Tuned index maintenance and partition-aware queries; reconciliation runtime -15%, deadlocks -15%.

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built transformer-based video summarization over 5,000+ recorded sessions; cut manual review 60-70%.
- AI-selected highlights aligned with human-curated moments ~85%.
- Implemented document Q&A with RAG; hallucinations ~30%, contextual accuracy >85%.
- Deployed as Flask API with a lightweight web interface for non-technical staff.

**PROJECTS**

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**Agentic Healthcare Claims Processing & Fraud Risk Intelligence System**

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

**Manga Lens - Multi-Provider AI Vision Chrome Extension**

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers) with per-provider payload handling. 29 sites supported across Manifest V3 service workers.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / OpenRouter

**Dream Decoder - Multimodal Creative Platform**

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. ~30% contextual alignment gain.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

**YOLOv8 Driver Drowsiness Detection**

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware.

PyTorch, YOLOv8, OpenCV, ONNX

**EDUCATION**

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**Master of Science, Computer Science - Kent State University**

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems.

**Bachelor of Technology, Computer Science - KL University**

Jun 2019 - May 2023

**CERTIFICATIONS**

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Deep Learning Specialization - DeepLearning.AI (Coursera)

2024